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Technical Report No. 52 Guidance for Good Distribution Practices (GDPs) For the Pharmaceutical Supply Chain

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Guidance for Good Distribution Practices (GDPs) For the Pharmaceutical Supply Chain

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Table of Contents

1.0 INTRODUCTION 2
1.1 Purpose and Scope 2
1.2 Responsibility 2

2.0 GLOSSARY OF TERMS 3

3.0 REQUIREMENTS 4
3.1 Stability 4
3.1.1 Storage Temperatures 4
3.1.2 Shipping Temperatures 4
3.1.3 Stability Testing to Support Distribution 5
3.2 Distribution Control Management 5
3.2.1 Qualification and Training of Personnel 5
3.2.2 Premises and Equipment 5
3.2.3 Material Handling 6
3.2.3.1 Receiving and Storage 6
3.2.3.2 Product Status 6
3.2.3.3 Controlled Drugs 7
3.2.3.4 Counterfeit Products 7
3.2.3.5 Records 7
3.2.4 Storage and Inventory Control 7
3.2.4.1 Inventory 7
3.2.4.2 Product Status Control 7
3.2.4.3 In-Transit Storage 8
3.2.5 Transportation 8
3.2.6 Product Disposition and Distribution 8
3.2.7 Product Protection 9
3.2.8 Returns Management 9
3.2.9 Exception Management 10
3.2.9.1 Corrective Action and Preventative Action (CAPA) System 10
3.2.9.2 Complaints 10
3.2.9.3 Quality Holds 10
3.2.9.4 Product Action 10
3.3 Performance Management 10
3.3.1 Performance Measurement and Reporting 11
3.3.2 Self-Inspection (Internal Audit) 11
3.3.3 Management Review Meetings 11
3.4 Supply Chain Partner Management 11
3.4.1 Partner Selection 11
3.4.2 Quality Audit 12
3.4.3 Quality Agreements 12
3.5 Business Review Meetings 12

4.0 APPENDIX 13
4.1 Good Storage and Shipping Practices Checklist 13

5.0 ADDITIONAL READING 31
5.1 Selected Global Laws and Regulations 31
5.2 Selected Publications 31
5.3 Selected Regulatory Authority Websites 31

FIGURES AND TABLES INDEX

Table 1.1 Seven Pillars of Good Distribution Practices 2

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1.0 Introduction

This technical report describes the overall quality system for distribution of pharmaceutical products and is meant to assist manufacturers in assuring that quality, integrity, and efficacy of the product are not compromised in the distribution channels.

The document summarizes both regulatory requirements and industry best practices. However country-specific regulations should be followed, as applicable.

1.1 Purpose and Scope

The purpose of this guidance is to provide direction for distribution practices, including handling, storage, transportation and distribution. This document is intended for use as guidance for the pharmaceutical industry and their supply-chain partners.

Good Distribution Practices (GDPs) can be divided into seven different categories, or pillars, as depicted in **Table 1.1** below. Each product should be evaluated against the pillars of GDPs to identify the individual categories that apply within each pillar. This Technical Report provides guidance on the first four pillars: Stability, Distribution Control Management, Performance Management, and Supply Chain Partner Management.

More specific guidance on the first pillar, Stability, is provided in *PDA Technical Report No. 53: Guidance for Industry: Stability Testing to Support Distribution of New Drug Products.* Separate PDA Technical Reports will provide guidance on the other GDP pillars: Qualification/Validation, Continuous Improvement, and Import/Export Compliance.

Table 1.1 Seven Pillars of Good Distribution Practices; this Technical Report covers the first 4

Stability	Distribution Control Management	Performance Management	Supplier Chain Partner Management	Qualification/ Validation	Continuous Improvement	Import/ Export Compliance
Storage Temperature Shipping Temperatures Stability Testing to Support Distribution	Qualification and Training of Personnel Premises and Equipment Material Handling Storage and Inventory Control Transportation Product Disposition and Distribution Product Protection Returns Management Exception Management	Performance Measurement and Reporting Self Inspections Management Review Meetings	Partner Selection Quality Audit Quality Agreements Business Review Meetings	Ambient Temperature Profiles Passive Shipping Systems Active Shipping Facility Qualification Warehouse Management System Validation Distribution Validation Master Plans	Industry Trends Regulatory Trends Requalification	Customs Release Documentation Control Product Tracking

1.2 Responsibility

This guidance applies to all parties that are involved in the handling, storage and/or transportation of regulated products, including, but not limited to, the manufacturer, the holder of the marketing authorization, third party Logistics Service Providers (termed 3PLs), wholesale distributors, and transportation carriers.

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2.0 Glossary of Terms

Change Control
The processes and procedures to manage changes.

Corrective Action and Preventative Action (CAPA)
Actions taken to eliminate the cause of an existing (corrective) or potential (preventive) non-conformity to prevent its recurrence.

First Expiration, First Out (FEFO)
A method of controlling inventory to ensure that the material with the shortest remaining shelf-life is distributed first.

ISO/IEC
International Organization for Standardization/ International Electrotechnical Commission.

Logistics Service Providers (LSP or 3PLs)
Freight forwarders and brokers that assist in moving the product, but may not physically touch the product.

Lost Work Day (LWD)
A day on which an employee does not work because of injury in a work-related accident.

Mean Kinetic Temperature (MKT)
International Conference on Harmonization (ICH) Q1A(R2) *Stability Testing of New Drug Substances and Products* defines MKT as "a single derived temperature, which, if maintained over a defined period, would afford the same thermal challenge to a pharmaceutical product as would have been experienced over a range of both higher and lower temperatures for an equivalent defined period" (uses the Arrhenius equation).

Root Cause Analysis (RCA)
An evaluation conducted for the purpose of identifying the cause of the deviation or non-conformance.

Transportation Service Providers (TSP)
Companies that physically transport the product.

Validation Master Plan (VMP)
The plan that documents rationale for the approach to validation and lists all systems and their validation status. (**Note:** The VMP can be used to document the rationale for number of monitors and revalidation frequency, as well as other system justifications.

Verified-Accredited Wholesale Distributors (VAWD)
An accreditation governed by the U.S. National Association of Boards of Pharmacy for pharmaceutical wholesale distribution facilities indicating that they are in compliance with state and federal laws.

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3.0 Requirements

Good Distribution Practices (GDPs) should include written procedures describing the different operations that may affect the quality of products and/or distribution activity. Refer to the pillars table, presented in **Section 1.1.**

3.1 Stability

Data from Stability studies is used to determine the temperature conditions required for product storage and transportation.

3.1.1 Storage Temperatures

When specific temperature conditions are required, storage or cargo areas should be equipped to maintain the appropriate temperature. Temperature should be monitored with calibrated temperature recorders or other devices that will indicate when the specific temperature range has not been maintained and for how long. Control and monitoring should be adequate to maintain all parts of the relevant area within the specified temperature range. Monitoring of temperature should be present in at least one of the following forms and should result in a disposition-based review of temperature data:

- Continuous monitoring
- Active, qualified monitoring system with automated alarming mechanisms to notify of temperature excursions as they occur.

A contingency plan between the manufacturer or distribution center and approved carrier should be implemented and documented to provide a consistent, effective response to foreseeable events.

Equipment used for monitoring should be calibrated at defined intervals.

All records should be made at the time each operation is done and in such a way that all significant activities or events are traceable.

3.1.2 Shipping Temperatures

Pharmaceutical products must be transported in a manner that ensures products will be maintained within an acceptable temperature range. Temperature should be monitored with calibrated recorders unless it is demonstrated through temperature mapping or use of qualified packaging containers that shipping conditions will not expose product to conditions outside of the shipping requirement. The acceptable temperature range may differ from the conditions specified for long-term storage and is determined by performing product temperature-excursion studies. These products may be shipped outside of their respective label storage conditions provided stability data or scientific/technical justification exists demonstrating that the product quality is not affected and that the data or justification meets the applicable national and international requirements. **Note:** some companies may choose to include shipping conditions on product labeling.

Appropriate stability data must be generated to prove that excursion temperatures have no negative impact on product quality. This data can be generated by accelerated and long term stability studies, forced degradation studies, or temperature cycle freeze-thaw studies. Stability data should be allocated into the various supply chain segments to support cumulative temperature excursions. If degradation profiles established from stability data meet the profiles predicted by the Arrhenius equation at established minimum and maximum temperatures (supported by actual stability data), then Mean Kinetic Temperature (MKT) may be an acceptable method for evaluating product impact. **Note:** it is necessary to ensure that the local regulatory agency allows the use of MKT.

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For products that are monitored during transportation, if temperature excursions occur outside of the labeled storage conditions or established shipping ranges, an investigation must take place to determine:

- the root cause of the temperature excursion
- any necessary corrective actions (if warranted)
- any potential product impact (must be documented)

Product impact should be addressed and documented for any temperature excursion and cumulative excursion impact during storage and distribution. Unplanned temperature excursions and planned system deviations should be properly documented and approved by the quality function.

3.1.3 Stability Testing to Support Distribution

Packaging should ensure protection against mechanical impact (damage) on product during handling and transport. The transportation modes and destination should be considered in determining the necessary packaging materials.

When specific temperature conditions are required, packaging used to control temperature should maintain the appropriate temperature range during handling, storage, transportation and distribution, and must be qualified. The methods used for qualifying packaging systems should be based on risk and the product's sensitivity to temperature change and must be documented (e.g., validation master plan). The result of the risk assessment will provide the rationale for the extent of testing and the need for ongoing temperature monitoring.

Each packaging/pack-out operation throughout distribution should be conducted under monitored conditions that are supported by stability data. Timelines for situations like Time Out of Refrigeration (TOR) should be defined, controlled and justified through appropriate stability data and in accordance with appropriate Service Level or Quality Agreements. Data must be available to support timeframes outside of labeled temperature and/or humidity.

3.2 Distribution Control Management

Distribution control management encompasses the processes and controls necessary to ensure product integrity and traceability are maintained from the point of receipt through product distribution.

3.2.1 Qualification and Training of Personnel

Procedures and conditions of employment, including background checks, should be designed and implemented to minimize the possibility of product diversion. SOPs should be in place for all job functions. An adequate number of warehouse and shipping personnel should be trained in the procedures for proper storage and distribution of each product type (hazardous, narcotic, etc.), and in the safe handling of product spills and other emergency procedures. A training program that includes both initial and ongoing training should be developed for each job function. The completion of training should be documented and training records should be maintained for each associate.

3.2.2 Premises and Equipment

Premises and equipment should be qualified as adequate to ensure proper management and distribution of products. A warehouse should be:

- Secure with controlled access and features to protect against invasion, adulteration and theft

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- Structurally sound, including hardened and sealed floors and no signs of leaks
- Sufficiently lit to allow accurate and safe operations
- Designed so that washrooms and eating areas are sufficiently separated from operational areas
- Clean and sufficiently maintained
- Constructed to protect against floods
- Free of agents that could cause infestation (includes the warehouse surroundings)
- In compliance with all local and federal/national regulatory requirements
- Environmentally controlled (temperature, humidity), where appropriate

3.2.3 Material Handling

Procedures for the delivery, receipt, storage and shipment of products are collectively referred to as material handling

3.2.3.1 Receiving and Storage

Receiving and shipping areas should:

- Protect deliveries from weather conditions or other elements that may have an adverse effect on materials/products during loading and unloading
- Be separate from storage areas

Deliveries should be examined at receipt in order to ensure:

- Materials/products are in good condition and not damaged
- Materials have been received from approved suppliers and within expiry.
- Product is in its original container, seals are intact and product is not suspected of being adulterated
- The identity and quantity of materials received correspond to the materials ordered

Products subject to special storage conditions should be:

- Identified with required storage conditions on the documentation (e.g., airway bill) and outside packaging
- Stored in accordance with specified instructions.

3.2.3.2 Product Status

Typically, commercial products should be approved and released for sale prior to being received in. It is acceptable to receive and/or to dispatch products in quarantine under the following conditions:

- Transfer of product is made between sites and/or manufacturer's third party warehouses where appropriate controls are in place to ensure that quarantined products will not be dispatched to customers prior to final release
- There is documented approval for quarantine shipments from the responsible quality function from both the receiving and shipping locations (e.g., quality agreement)
- There is a warehouse management system that ensures:
  - A shipment will have an identification such as "quarantined product"
  - The status of a product is accurately assigned
  - The authority and access to perform the status change will be defined by the system tools, procedures and training.

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3.2.3.3 Controlled Drugs

Local regulatory requirements should be followed for the activities and documentation involving controlled drugs (e.g., importing, warehousing, transport, handling, records and official reports, as applicable).

For importing or exporting activities, both the distribution center and receiving country are responsible to communicate and implement the requirements.

3.2.3.4 Counterfeit Products

Counterfeit products found in the distribution network should be kept apart from other saleable products in a restricted and controlled area to avoid any confusion. They should be clearly labeled as not for sale. Competent authorities and the holder of the marketing authorization of the original product should be informed immediately. Counterfeit products should be dealt with in accordance with federal/national or local regulations.

3.2.3.5 Records

Records should be generated at the time each operation is completed and in such a way that all significant activities or events are traceable. Records should be understandable and readily available. Records retention schedules or equivalent documents should be established and maintained according to applicable record retention regulations.

3.2.4 Storage and Inventory Control

Computerized systems and/or written procedures should be implemented to record and track product information that is critical to protect products and customers. This information includes lot numbers, expiration dates, storage locations, product status and any other information required to track the product from receipt through distribution and to ensure that customers receive quality product. Systems/procedures should address the following:

3.2.4.1 Inventory

There should be a documented system for the recording and traceability of lot numbers and expiration dates. Inventory should be periodically reviewed to assure the control of identity, quantity, location and status of each inventory item.

Computerized systems should be appropriately qualified and validated.

Inventory accuracy must be maintained and verified. Periodic cycle counts or annual inventory reviews should be performed and documented in order to detect and investigate discrepancies and identify corrective actions, and should be used as tools for improvements to inventory management.

3.2.4.2 Product Status Control

There should be a validated system to control and maintain product status (e.g., on test, approved, quarantined, restricted). The system should:

- Be under appropriate change control
- Allow changes to product status to be made only by appropriate quality personnel.

Computerized systems should be appropriately qualified and validated.

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3.2.4.3 In-Transit Storage

In-transit storage facilities (e.g., cross-docking center, airplane, airport, distribution hub and other facilities which are used to move product from one vehicle to another vehicle) may be used. All in-transit storage facilities should have adequate security and environmental control, and must be in compliance with all applicable laws and regulations prior to use.

3.2.5 Transportation

Products should be transported in such a way that product:

- Maintains its identification
- Does not contaminate and is not contaminated by other products or materials
- Is secure (i.e., protected against spillage, breakage or theft)
- Is not subjected to unacceptable changes in heat, cold, light, moisture or other adverse influences
- Is not subjected to attack by pests.

Products requiring controlled temperature storage should be transported with appropriate controls and monitored. Shipment time should not exceed the timeframe established through validation studies. Excursions may be justified through continuous temperature monitoring and appropriate stability studies.

Freezers used for storage of coolants such as chilled or frozen gel packs and the process for control of those coolants should be qualified. Vehicles containers and/or packaging should be appropriately matched to avoid negative effects on the integrity and quality of pharmaceutical products.

Products that are temperature controlled that ship internationally should be labeled per IATA Perishable Cargo Regulations.

Vehicles used to distribute, store or handle pharmaceutical products should be suitable for their use and appropriately equipped to prevent exposure of the products to environmental conditions that could affect their stability and packaging integrity, and prevent contamination of any kind. Carrier studies may be used to support adequacy and qualification of equipment in combination with active load performance qualification studies. Monitoring should take place in at least the front and back of vehicle and should support duration of shipment.

Vehicles should be inspected for cleanliness and temperature setting, if applicable, prior to loading and transporting product. This inspection should be documented.

3.2.6 Product Disposition and Distribution

Product can only be distributed to authorized parties or to persons authorized to supply and/or receive products. Products due to expire first should be distributed first (FEFO). Adequate controls should be in place to prevent the distribution of expired products throughout the supply chain.

For pharmaceutical products supplied to a person authorized or entitled to supply products to the public, a document should be enclosed that makes it possible to ascertain the date, the name and pharmaceutical form of the product, the quantity supplied, and the name and address of the supplier (e.g., bill of lading, purchase order or invoice). Documents should also contain quantity shipped and lot number of material.

It is critical to coordinate shipments throughout the supply chain to ensure proper timing for products

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to be transported and received with regard to holidays, weekends or other forms of interruption, such as weather-related delivery delays or customer receipt schedules. Any potential customs-related delays, including those by the U.S. FDA, U.S. Department of Agriculture, health ministries in other countries, or any other governmental agency resulting from international shipments, must also be considered.

3.2.7 Product Protection

Fundamental to the outcomes of a Quality System surrounding GDP is the need to ensure that products are handled, stored, transferred and transported in a manner that does not negatively impact or degrade the product. Thus, GDPs should protect the product from external environmental conditions that might cause undue deterioration. These include chemicals, light, moisture, temperature, pest infestation, or other adverse influences that could cause microbial and/or chemical contamination. If a pesticide program is in place the pesticide should be registered as required by local regulations.

Paths to degradation of product should be established through stability testing, and required shipping conditions defined and communicated to the parties of the supply chain so as to enact or implement appropriate means of environmental control. Data, such as filed stability test results, forced degradation studies and temperature cycling studies, may be used to support supply chain temperature excursions.

The storage facilities and vehicles or containers used to transport product should be clean and free from litter, dust and pests. Adequate precautions should be taken against spillage or breakage that attracts microorganisms, causes cross-contamination, or could create a health or safety risk to personnel. There should be adequate space for the orderly placement of materials to prevent mix-ups between drug products to prevent contamination.

Products with broken seals or damaged packaging or suspected of possible contamination should be withdrawn from saleable stock, and if not immediately destroyed, they should be kept separate, secure and quarantined so that they cannot be sold in error or contaminate other goods.

Sufficient security should be provided to prevent theft and product diversion.

3.2.8 Returns Management

Any shipped product presented to a storage facility for return must be evaluated in accordance with return procedures to determine if the material must be destroyed or if it is eligible for restocking. Returns should be examined upon receipt, as described in **Section 3.7.** Returned products should only be evaluated for restocking eligibility if:

- They are in original, unopened containers, in good condition, not damaged, and it is known and documented that they have been stored and handled under proper conditions
- They have been evaluated by an authorized person
- There is documentation that required temperatures were maintained and that security of the materials was never compromised
- Documentation indicates the reason for return and that the product was not received into stock by the customer so that chain of custody is maintained. **Note:** many companies do not receive cold chain back into stock due in part to inconsistencies in controls for the return shipment.

Returns for destruction should be processed by an authorized pharmaceutical destruction company

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unless the manufacturer has the means to destroy the goods.

3.2.9 Exception Management

Any excursion from the defined process, as identified by the manufacturer, constitutes an exception. An exception management system should be put in place to track all non-conformances and the investigation of the non-conformances so that product and quality problems may be routinely analyzed. The investigation into a quality problem or nonconforming product should be conducted to a level commensurate with the significance or risk of the problem.

The exception management system should be documented. The documentation should identify the activities that should be treated as non-conformances and a description of the steps and documentation that is required to complete the investigation, including product impact. A targeted timeframe for closure should be defined along with a schedule for review with management.

3.2.9.1 Corrective Action and Preventative Action (CAPA) System

A system with documented procedures should be developed including the following elements:

- Description of the investigation that took place
- Root cause, if one exists
- Description of how root cause was identified
- Corrective actions and preventative actions
- Owners and target due dates for corrective actions and preventative actions.
- Verification that the corrective and/or preventative action(s) were implemented and found effective.
- Evidence of management review

The system should define a targeted timeframe for the identification of root cause and the closure of the associated action plans. Additionally, CAPAs should be included in management reviews.

3.2.9.2 Complaints

Procedures for receiving, reviewing, evaluating and investigating complaints should be established, documented and maintained. The complaints should be processed in a uniform and timely manner. History of product complaints should be maintained to track and trend complaints and responses and any resulting CAPAs.

3.2.9.3 Quality Holds

A system, including documented procedures, should be in place to implement holds on products, as necessary, to prevent them from being distributed to customers. The responsibility and ability of affecting the quality release status of a product should be restricted to personnel who are properly trained and authorized by the quality function.

3.2.9.4 Product Action

A system, including documented procedures, should be in place to respond to all product actions, including recalls, withdrawals and corrections. The system should include a requirement to segregate and secure product requiring action to prevent distribution to customers.

3.3 Performance Management

Methods and procedures should be established to review the performance of the quality system

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on a regularly scheduled basis and to identify opportunities for continuous improvement. These methods and procedures should include Performance Measurement and Reporting, Self-Inspection (Internal Audit), and Management Review Meetings.

3.3.1 Performance Measurement and Reporting

Performance of the components of the quality system should be measured against pre-defined goals to gauge performance and to identify opportunities for improvement. Reports should be generated and reviewed on a regularly scheduled basis to track performance, monitor trends, and identify areas for improvement.

3.3.2 Self-Inspection (Internal Audit)

The quality assurance program should require self-inspections, which should include the quality systems of internal supply chain partners. These self-inspections should be conducted in order to monitor the implementation of GDP and compliance with GDP principles, as well as to trigger necessary CAPA measures and monitor CAPA effectiveness to achieve continuous improvement. This is termed a 'closed loop supply chain,' in which the manufacturer is aware of their supply chain partners and operations.

3.3.3 Management Review Meetings

Quality management system reviews should take place on a regularly scheduled basis and should cover:

- Audit results
- Performance metrics (actual vs. goals)
- Status of non-conformances, corrective actions and preventative actions
- Customer feedback (returns, complaints, etc.)
- Product status (recalls, non-availability)
- Changes that may impact the quality systems.
- Opportunities and recommendations for improvement

3.4 Supply Chain Partner Management

The quality of supply chain partners is critical to product safety. Methods and procedures should be implemented to provide a consistent and repeatable process for selecting, maintaining and evaluating the performance of supply chain partners. These procedures should address the following:

3.4.1 Partner Selection

There should be a process in place to evaluate and select service providers that will be able to support GDP compliance. The process for selection of a service provider should include satisfactory findings in areas that are critical to the portion of the distribution system that their service applies (see Good Storage and Shipping Practices Checklist). These include but are not limited to:

- Transportation provider's safety records
- Regulatory certifications (e.g. verified-accredited wholesale distributors [VAWD], ISO, etc)
- Operational Policies and Procedures
- Vehicle and Equipment (to include refrigerated trucks, sea containers, aircrafts, etc)
- Security

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3.4.2 Quality Audit

External inspection should verify that written agreements have been executed with the service providers and that the service providers adhere to the terms and conditions of the agreements (e.g., master service agreements, quality agreements, etc). Standards should be established to include agreement on metrics for regular business reviews where metrics are reviewed and errors and continuous improvements are discussed. Quality system reviews and audits (via on-site visits or via questionnaire, depending on the risk) should be performed by the quality organization from the manufacturer or the marketing authorization holder, and appropriately documented to allow for reporting, and follow-up.

The attached audit checklist (see **4.0 Appendix**) covers all sections of the pharmaceutical supply chain. All sections may not be applicable to specific services providers, however if a section is not applicable (N/A) justification should be provided. For example, warehousing of commodities like expanded polystyrene and corrugate do not need temperature profiles.

3.4.3 Quality Agreements

It is the responsibility of the manufacturer and/or holder of the marketing authorization to ensure that the different aspects of this guideline are appropriately assigned and implemented to the different parties of the supply chain. The distribution of responsibilities should be formalized in covenants or "Quality Agreements" with all third parties. Such agreements should be periodically updated.

3.5 Business Review Meetings

Business review meetings should take place on a regularly scheduled basis and should cover:

- Results of quality audits
- Performance metrics (actual vs. goals)
- Status of corrective actions and preventative actions.
- Product availability (recalls, returns, non-availability)
- New product requirements (storage and transportation)
- Changes that may impact products or the quality systems
- Opportunities and recommendations for improvement

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4.0 Appendix

4.1 Good Storage and Shipping Practices Checklist

The following is an assessment tool that may be used in the evaluation of pharmaceutical and/or biotech distribution centers, wholesalers, 3rd party distributors, related vendors, service providers, etc. This checklist is intended to facilitate assessments and/or audits in the area of Good Distribution Practices.

Company Name

Facility Address: _____

Facility Board of Pharmacy Registry Number (BoP): _____

Facility Drug Enforcement Agency (DEA) Registry Number (as applicable): _____

Person (s) Answering Questionnaire and Position (Site): _____

Auditor (s): _____

(circle one)

The site is a: Distributor/Wholesaler or 3rd Party Distributor/Vendor or Service Provider

A. Accreditation and Licensure	Comments
1. What accreditations/licenses are held (ISO, cGMP, GDP, BoP, VAWD, etc.)	
2. By whom and when were the last external Inspections or Audits conducted (by BoP, Customer, FDA, DEA, etc)	
3. What system is used to maintain licensure?	

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B. Quality System Comments	Comments
1. Do you have a separate and distinct independent Quality organization?	
2. Does your organization employ quality by design and product development?	
3. Does your organization have a formal review process for: • quality policy and objectives • results of audits and other assessments • customer feedback and complaints • analysis of data trending results • status review of actions taken to correct or prevent problems from reoccurring (CAPA) • follow-ups from the previous management review meeting • changes in the business that may impact quality systems • product characteristics or business processes • meeting customers' needs	
4. Are the organization and resources sufficient to support the business and quality systems?	
5. Are outsourced operations used and how are they qualified to perform their service?	
6. Do you have written procedures that provide control and maintain operations?	
7. Do you have a calibration program?	
8. How are allowable shipping ranges established?	
9. Do you have a qualification program?	
10. Do you have a deviation and investigation program?	
11. Do you have a corrective and preventative action (CAPA) program?	
12. Do you have a formalized, documented training program?	
13. Do you have an audit program?	
14. Do you have a periodic assessment of your temperature-controlled processes?	
15. Do you have a change control program?	
16. Do you have management controls in place?	
17. Do you have tracing, tracking and proper audit trails such that faulty product could be recalled?	

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I. FACILITIES	
C. Building and Housekeeping	**Comments**
1. Is there a scheduled housekeeping program?	
2. Is the storage area maintained free of dust, debris, or extraneous material?	
3. Are the floors sealed to minimize dust and facilitate cleaning?	
4. Are there written procedures describing cleaning frequency, methods, and materials to be used in cleaning the building?	
5. Are there cleaning logs maintained in which cleaning activities are documented per cGMP standards?	
6. Are the cleaning supplies and equipment identified and segregated from products and commodities?	
7. Are there written procedures that define special handling precautions for the storage and handling of hazardous materials and substances?	
8. Are the washrooms or locker areas clean and properly maintained?	
9. Are there indications of roof or sprinkler system leaks?	
10. Is the height of the floor and drainage above any reasonable potential flood level?	
11. Is lighting sufficient to allow all operations (e.g., stock picking) to be accurately and safely conducted?	
12. Are doors designed to provide security for the product and restrict entry of vermin?	
13. Are skylights or trap doors lockable to prevent theft or unauthorized entry?	
14. Are there adequate washrooms or locker areas that are isolated from the main storage areas?	
15. Is there a segregated area for the holding and storage of returned/rejected goods?	
16. Is there a separated, designated area for the repackaging of finished goods?	
17. Is there back-up power in the event of a power failure?	
18. How long does the back-up power last?	
19. Are the back-up generators on a regular maintenance and testing schedule?	
20. Is the perimeter of the site fenced and fully enclosed to limit unauthorized access?	
21. Is there an automated alarm system to a central station in the event that environmental conditions are not being met?	
22. Is the maintenance performed by in-house staff?	
23. How are maintenance requirements managed and scheduled?	

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D. Pest Control	Comments
1. Is there a regular pest control and rodenticide program?	
2. Are written service records available?	
3. Is there a written pest control procedure?	
4. Is there evidence of rodent, bird, or insect infestation?	
5. Are the pesticides used on the EPA/OSHA/USDA approved list?	
6. Are bait boxes used within the facility?	
7. Are electronic flying insect killers present?	
8. Are bait boxes and/or electronic flying insect killers inspected and cleaned on a routine basis (at least once a month)?	
9. Are the inspections/cleanings (bait boxes and/or electronic flying insect killers) documented?	
10. Is a diagram maintained and available at the facility that shows the location of bait boxes and/or electronic flying insect killers to ensure that all are serviced, cleaned, and inspected per schedule?	
11. Is action taken to prevent entry of birds?	

II. PRODUCT CONTROL

E. Product Storage	Comments
1. Are products stored off the floor on pallets/racks?	
2. Are the products stored along the outer walls of the facility maintaining a separation from the wall of 6" to 18" (Inspection Perimeter)?	
3. Are chemicals/hazardous materials stored in approved storage cabinets and separated from product storage?	
4. Are products subject to insect infestation stored separately from general product storage area, i.e., bagged foods such as flour, peanuts, grain, etc.?	
5. Does the system assure that stock cannot be inadvertently transferred to other companies using the facility?	
6. Are products and bulk materials properly stacked on pallets to avoid damage, i.e., where pallets are stacked are corrugate interleaves used to protect the product?	
7. Are products physically segregated to prevent mixing of separate products and/or batch numbers?	

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E. Product Storage (continued)	**Comments**
8. Are outdated and damaged products removed from general storage and segregated?	
9. Are controlled drugs stored in the facility?	
10. If the response to item 9 above is yes, is a separated and secured area of the facility provided for controlled drug storage and control?	
11. Has the Facility been inspected for DEA compliance if applicable?	
12. Are there written instructions describing storage procedures, materials handling and documentation requirements?	
13. Are there written procedures indicating recommended storage conditions, precautions to be observed, and shelf life for each stored product?	
14. Upon receipt, is the supplier's documentation checked against purchase order to verify the correct label description, product number, quantity, batch number, and physical condition?	
15. Are the records of each delivery retained? Do they include the description of the products, presentation, quantity, supplier, supplier's lot number, date of receipt, expiration date and any temperature requirements during transport?	
16. If controlled drugs or high value products/goods are stored in the warehouse, are there special procedures available defining product handling, authorized personnel, identification requirements and access to the secured areas?	
17. Are there systems in place to ensure that temperature sensitive products are received, transported, and stored under the appropriate temperature conditions? Is there documented evidence to support the actual storage conditions?	
18. Is access to the storage areas limited to only authorized personnel?	

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F. Temperature Control/Calibration Storage	Comments
1. Is the general warehouse temperature controlled within a range to meet label claim? (i.e. 15°C to 25°C)	
2. Are products requiring lower temperature conditions stored in a suitable storage area (cool room or refrigerator)?	
3. Are all controlled temperature storage areas equipped with calibrated recorders/devices that indicate when required temperature or relative humidity ranges have not been maintained?	
4. Are temperature-recording devices calibrated (at least once a year) against acceptable standards? (i.e. NIST or ISO/IEC 17205)?	
5. Are calibration records available for each temperature-recording device?	
6. Are recorders in ambient, cold chain and freezer storage units furnished with low/high temperature alarms?	
7. Are the alarms set such that it would permit timely corrective action in the event temperatures were trending out of range?	
8. Is the temperature in each controlled temperature storage area monitored on a regular basis?	
9. Are temperature records maintained and are they signed and dated?	
10. Is there a written procedure describing the action(s) to be taken when there is a temperature excursion?	
11. Have controlled temperature areas been temperature mapped to determine high and low temperature areas within the storage area during seasonal extremes?	
12. Were the temperature mapping studies conducted during the height of business activities for a sufficient period of time (at least 3 days)?	
13. Is there a program in place to review temperature mapping as part of your validation plan every three years to identify if re-mapping is necessary?	
14. Is there a regular preventative maintenance (PM) schedule in place to maintain equipment?	
15. Is there a documented notification process if equipment is found to be out of calibration?	
16. Are systems maintained in the event of a power failure?	
17. Are there written procedures that describe how temperature records are to be maintained for all of the temperature controlled areas (operation and maintenance)?	
18. Has Mean Kinetic Temperature (MKT) been determined for each of the storage areas?	

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F. Temperature Control/Calibration Storage (continued)	**Comments**
19. Are a suitable number of recording devices installed to record temperatures?	
20. Are alarms tested for functionality on a regular and documented basis?	
21. Is there a contingency plan in place in the event of catastrophic failure of equipment or emergency (i.e. Business Continuity Plan)	
22. Is there a pallet management program in place to ensure appropriate handling and treatment of wood pallets? (It is recommended that appropriate moisture specification be in place.)	
G. Temperature Control Distribution	**Comments**
1. Is for each temperature-recording device a calibration certificate or validation certificate supported with manufacturer's data available?	
2. Are systems and process in place to assure the correct setting and configuration of the temperature-recording devices?	
3. Are a suitable number of recording devices installed to record temperatures at hot/cold spots of the used shipping container/carrier?	
4. Are data-loggers equipped with downloading capabilities (i.e. USB interface, pdf-datalogger) and/or are all recipients trained and equipped to download, handle and interpret devices?	
5. Are systems and processes in place to prevent manipulation of temperature records?	
6. Are automated processes and/or systems in place to interpret the collected data against the available stability data (i.e. multi-level-alarm zones) or is a manual evaluation process established?	
7. In case of deviation, are tools and processes available to return data safely to the investigating department in order to assure a timely analysis of the raw data.	
8. Is a CAPA system in place which involves all relevant parties including the supplier of the temperature-recording devices?	
9. For multiple-use devices is a regular maintenance and calibration program and/or an exit calibration installed?	

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H. Inventory Control	Comments
1. Is there a first expiration, first out (FEFO) stock system?	
2. Are there written procedures that define the responsibilities for the preparation, communication, and implementation of QA Hold (Nonconforming product) directives and dispositions?	
3. Is nonconforming product marked or systemically segregated as blocked or unavailable?	
4. Is the disposition of nonconforming product documented and traceable to destruction?	
5. Are all materials under a stock control system?	
6. Does the stock control system record batch or lot numbers and expiration dates, where required?	
7. Are computer based stock control systems that affect lot traceability validated?	
8. Are periodic cycle counts conducted and discrepancies noted and investigated?	
9. What threshold is used for conducting an investigation into a cycle count discrepancy?	
10. Are cycle counts performed daily for all pick-from locations within your warehouse? If no, please identify frequency and schedule followed for conducting cycle counts.	
11. Is the stock periodically inspected for obsolete, expired and damaged materials?	
12. Are precautions taken to prevent the issue of outdated or short dated stock? Is the data in inventory system challenged or audited for accuracy against inventory on the shop floor?	
13. Are distribution and recall records readily accessible and maintained? Are procedures in place to prevent restock of recalled materials?	
14. For temperature-controlled products, is inventory moved from receiving docks to their manufacturer-designated storage environment as quickly as possible? Is the put-away process proceduralized and supported by stability data?	
15. For temperature-controlled products, is any time spent in uncontrolled environments limited to ensure product integrity?	
16. In the event that a deviation has been observed by the storage site upon receipt, is the product in question placed in quarantine status?	
17. Is there a documented procedure which defines subsequent actions, related to #16 above?	
18. Are procedures in place to prevent distribution of products returned due to customer complaints?	

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I. Product Distribution	Comments
1. Are stock distribution records maintained by lot number?	
2. Are there written procedures for dispatch and preparation of shipping orders?	
3. Do procedures include instruction for shipping the different types of temperature sensitive products?	
4. Are the shipping containers adequate to provide product protection?	
5. Where temperature control is needed, have the shipping containers been validated for temperature and time?	
6. Are containers used for shipment of temperature, moisture or light sensitive products properly labeled to indicate handling, storage and transport requirements?	
7. For temperature controlled products, are the vehicles used for distribution suitably equipped to ensure products are maintained at the correct temperature?	
8. Have the temperature-controlled vehicles been qualified for the temperature range(s) or is continuous temperature monitoring of every shipment applied? Provide details on how vehicles are qualified or on how the continuous monitoring system is validated.	
9. If shipping of unreleased product is part of the quality system are controls in place to ensure product is not distributed (e.g. QA Hold) until final QA testing is complete?	
10. Are distribution records maintained that will permit prompt recall, market withdrawal/correction, or lot withdrawal, if required?	
11. Are there written procedures in place to comply with in the event of a quality-related product action (e.g., recall, market withdrawal/correction, lot withdrawal)?	
12. What steps are taken to assure that receiving locations are made aware of special handling and storage specifications?	
13. Is the distribution of medical samples and promotional materials done according to the same control procedures and GMP & GDP principles which are applicable to the distribution of commercial product presentations?	

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J. Systems	Comments
1. What systems platforms and warehouse management system does the warehouse have?	
2. What interfaces would be required to release orders and confirm shipments?	
3. Does the system provide for lot-controlled storage?	
4. Does the storage, picking and distribution system control using the lot number?	
5. Does the system have a list/lot/customer ship to, and quantity in the event of lot recalls?	
6. Does the system provide for QA holds by lot number with allocation and picking and shipping blocks?	
7. Can the system handle differing label codes of the same lot?	
8. Can the system identify short dated product?	
9. Can the system pick based on first expiration dates?	
10. Can the system pick based on lot number specific I.D.?	
11. Can the system specify one lot per customer?	
12. Does the system provide for put away logic with fast moves near the shipping zones?	
13. Does the system handle returned goods?	
14. Does the system have security for QA hold release?	
15. Will the system recognize temperature-controlled product and provide for systems directed put away in cool room area?	
16. Does the same logic apply for hazardous materials?	
17. Does the system support cycle counting?	
18. Does the system provide for one lot/list number per location?	
19. Does the system provide for bar-coding applications?	
20. Does the warehouse have RF technology and scanning capabilities?	
21. Have there been any break-ins, thefts or products diverted in the last year? Provide details and actions taken as the result of such events.	
22. Is there active monitoring for suspicious orders of controlled substances per DEA regulations?	
23. Does the system have a disaster recovery program?	

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K. Insulated Shipping Systems (ISS) Transportation Temperature Controls	Comments
1. How were insulated shipping systems chosen?	
2. How were the systems challenged?	
3. How were temperature profiles developed?	
4. Were they tested against seasonal extremes?	
5. Were they tested against various modes of transport?	
6. Were they tested against routing supply chain routes?	
7. Are shipments monitored during transport?	
8. Are the insulated shipping containers periodically tested to evaluate performance during actual shipments?	
9. How were conditioning times determined for refrigerants?	
10. Are there written specifications for the components used in temperature controlled shipments (passive and active)?	
11. Are components inspected at receipt?	
12. How are changes to the components managed?	
13. Are there written procedures documented and followed that detail the assembly and position of product, refrigerant, monitors and cushioning?	
14. How were the allowable quantities or volumes per container established?	
15. How were the protective requirements (i.e. shock & vibration) tested?	
16. What type of consumable coolant do your shippers use (dry ice, phase change material, etc)?	
17. What is the process for managing dry ice in the facility?	
18. How long can dry ice be stored prior to use?	
19. How is the storage of dry ice prior to use controlled?	
20. How is dry ice replenished during transit?	
21. What is the contingency plan if issues occur during transit?	
22. How is the process controlled during transit (with handoffs and partners)?	
23. Are there measures in place to minimize exposure to ambient conditions outside of the container specifications?	
24. Are time limits in place and tracked for transportation to ensure established hold times of insulated shipping systems are not exceeded?	
25. After insulated shippers are qualified is there a monitoring program to verify temperatures?	
26. Have there been any failures of the qualified insulated shipping system? If yes, provide details on how failures were addressed to include RCA and CAPA.	
27. Are transportation routes documented and controlled?	

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L. Carrier Transportation Temperature Controls	Comments
1. Are Service Agreements in place with each of the carriers that detail specific service requirements and transportation routes and carriers?	
2. Have the carriers been audited and assessed for their ability to provide services required?	
3. Are sufficient controls in place to assure that the supply chain is safe and secure?	
4. Are there processes in place to allow for accurate communication between the carrier and the driver?	
5. Are transportation vendors evaluated and certified by a documented process to permit them to be used as a carrier?	
6. Are transportation routes certified? If so by who? How are temperatures for a particular carrier, lane, mode and service level verified? (e.g. temperature mapping, or other method?)	
7. Does the vehicle possess continuous temperature monitoring equipment?	
8. Are subcontractors used to transport product?	
9. How are subcontractors advised of the requirements and how are they monitored for compliance?	
10. Have the transport vehicles been temperature mapped?	
11. Do temperature monitors accompany each shipment during transit?	
12. If monitors are not used how is temperature compliance assured?	
13. Are vehicles preconditioned to temperature prior to the start of loading?	
14. Are vehicle temperatures checked upon receipt?	
15. Are the cooling/heating devices on the trailer calibrated?	
16. Are the temperature monitoring devices on the trailer calibrated?	
17. Are alarm settings used to alert the driver or central station of a deviation or excursion?	
18. Is the vehicle equipped with a pre-trip diagnostic test at start up?	
19. Are seals used to secure the trailer?	
20. Are seal numbers recorded, documented on shipping documents and verified at end of trip?	
21. Are trailers equipped with global positioning devices for active, real time, location and temperature tracking?	
22. Are security and notification procedures in place in the event of product diversion or theft?	
23. Are vehicles inspected prior to loading to assure that no prior material or opportunities for contamination are present?	

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L. Carrier Transportation Temperature Controls (continued)	**Comments**
24. Are procedures in place to manage product requirements in the event of an equipment failure or excursion?	
25. Are loading patterns defined and documented?	
26. Are written procedures in place that specify the location and placement of monitors during transport?	
27. Are vehicles required to be fueled to a level sufficient to permit transport to destination prior to loading?	
28. Are procedures in place to assure that drivers' credentials are adequately checked and verified to be agents of the transportation company prior to loading?	
29. Are team or solo drivers used to transport high value, temperature sensitive loads?	
30. Are procedures in place to maintain proper custody and security at stopping points during transport?	
31. How are temperature and handling requirements communicated?	
M. Customer Returns	**Comments**
1. Are all customer returns inspected by the appropriate Quality Assurance unit?	
2. Does the Quality Assurance procedure include inspection of product to confirm it is in its original condition, there is no visible damage, correct quantity is present, no broken seals, no foreign matter or defective containers, case seals are present, and temperature maintained during distribution?	
3. What criteria are used for placing returns back into stock? Please specify if criteria includes (a) verifying chain of custody, (b) environmental conditions products may have been exposed to and (c) any restrictions in place for material that has left your chain of custody and control (ex. Material placed into stock by a customer)	
4. Are customer returns physically segregated from approved warehouse stock to prevent mix-up?	
5. Are returns reconciled to the sale of the product to the original customer, and proper credit given?	
6. Is there a documented system in place to assure Quality Assurance review and approval of redressing, restock or destruction of finished goods?	

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N. Complaints	Comments
1. Are there separate procedures (details) for addressing complaints for distribution or the product/product packaging?	
2. Are there separate procedures for addressing customer complaints related to temperature excursions?	
3. Are complaints documented, tracked, trended and investigated?	
4. Is there an escalation process in place for complaints identified as a significant issue or trend?	
5. Is corrective action, preventive action performed and verified effective as appropriate?	
O. Counterfeit & Tampering	**Comments**
1. Are there procedures addressing counterfeit and tampering or suspect counterfeit products?	
2. Are suspect counterfeit and tampered products isolated and distinguished from other pharmaceutical products?	
3. Do procedures outline a process for notifying the marketing authorization holder and regulatory authorities?	
4. Do procedures outline the return or destruction of confirmed counterfeited or tampered pharmaceutical products?	
5. Is there a process in place for identifying suspicious orders (unusual quantities or order patterns)?	
P. Material/Product Description	**Comments**
1. Are materials/products for destruction (e.g., damaged, outdated, expired) handled and destroyed in compliance with local and country regulations?	
2. How is destruction handled (e.g., third-party or in-house)?	
2a. If third-party, have they been audited?	
2b. If third-party, are records available for review?	
2c. If third-party, has a periodic review occurred to evaluate and demonstrate a proper chain of custody and audit trail?	
2d. Are records of the destruction maintained?	
3. Is there periodic and timely purging of material to be destroyed (e.g., not stockpiled)?	

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II. PRODUCT CONTROL	
Q. General	**Comments**
1. Is there a program for employees to maintain a good standard of personal hygiene?	
2. Are personnel under control of competent supervisor/ managers who understand GDP and GMP requirements?	
3. Are there designated eating and food storage areas separate from operations?	
4. Is access to areas limited and restricted to prevent entry by unauthorized personnel?	
5. Are pre-employment background checks conducted?	
6. Are employees required to pass drug testing prior to hire?	
7. Are employees in a random drug testing program?	
R. Training	**Comments**
1. Is there a documented training record for each employee?	
2. Does the training program include the training of contract/ temporary personnel, where applicable?	
3. Have personnel received the training necessary to perform their assigned duties and responsibilities?	
4. Are training records available for all applicable personnel (including contract/ temporary personnel, if applicable)?	
5. Is there a requalification program for personnel? Is the program maintained on an annual basis?	
6. Do employees receive retraining on revised documents? How is it documented?	
7. Are employees trained on how to review environmental monitoring? (Security, Operations, Maintenance) ?	
8. Are employees trained to identify atypical events or temperature excursions?	
9. Are employees trained on how to properly respond to alarms (Security, Operations, and Maintenance) ?	
10. Is there ongoing periodic cGMP training in place and documented?	
11. Is there a training curriculum for each position and each employee?	
12. How are trainers qualified to train on subjects?	
13. Are training records readily available/retrievable?	
14. Do associates have knowledge and are they able to regularly review training requirements in order to remain current and up to date in their training?	
15. Are contractors or subcontractors required to complete the same training as regular staff?	
16. How is contractor training documented and maintained?	
17. Are there positions or tasks that require certification (i.e. safe fork truck driving) prior to performing the task? If so, what are they?	

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S. Safety	Comments
1. Is there a Safety Program and Manager associated with the site?	
2. Does the site have any governmental safety inspections and licenses?	
3. Does the site have adequate sprinkler or fire water reserves, i.e.. Fire hydrants and portable extinguishers?	
4. Are the Fire extinguishers tested per manufacturer's specs every six months regarding use?	
5. Are employees trained on fire prevention and procedures?	
6. What is the accident rate of the warehouse?	
7. What are the requirements to drive a fork truck or use material handling equipment?	
8. Describe your safety program?	
9. How are accident rates monitored?	
10. How are accident rates communicated to employees?	
11. What incentives are offered associated with safety metric outcomes?	
12. What is the course of action if a fork truck or driver has an accident?	
13. Is there a requalification or recertification program if a driver has an accident?	
14. Is there a hazardous spill team at the site?	
15. How are hazardous spills managed?	
16. Is there a first aid or first responder team on site?	
17. Is there a lock out/tag out program on site	
18. Who is trained in lock out/tag out procedures?	
19. How is safety training handled for contractors?	
20. What is your OSHA recordable and lost work day (LWD) rate?	

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OBSERVATIONS AND RECOMMENDATIONS
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Site Evaluation

Approved: _____

Date Recieved: _____

Summary Report of Findings– Date available (30 days from audit date): _____

Signed: _____ Date: _____

Signed: _____ Date:

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5.0 ADDITIONAL READING

5.1 Selected Global Laws and Regulations

1. Annex 5, Good Distribution Practices (GDP) for Pharmaceutical Products, World Health Organization, 2005.
2. Guidelines for Temperature Control of Drug Products during Storage and Transportation (GUIDE-0069), Health Canada, 2005.
3. Guide to Control and Monitoring of Storage and Transportation Temperature Conditions for Medicinal Products and Active Substances, Irish Medicines Board, 2006.
4. Guidelines on Good Distribution Practice of Medicinal Products for Human Use (94/C 63/03), European Union, 1994.
5. Quality Guideline Q8: *Pharmaceutical Development;* International Conference on Harmonisation: 2006.
6. Quality Guideline Q9: *Quality Risk Management;* International Conference on Harmonisation: 2006.
7. Quality Guideline Q10: *Pharmaceutical Quality System;* International Conference on Harmonisation: 2007.
8. Federal Food, Drug and Cosmetic Act. Chapter V: Drug and Devices, Subchapter A, Sec 501 (a)(2)(B).
9. Federal Food, Drug and Cosmetic Act. Chapter III: Prohibitive Acts and Penalties, Subchapter A, Sec 301 (a)(b)(c).
10. Food and Drugs. *Code of Federal Regulations,* Part 211.
11. Perishable Cargo Regulation. International Air Transport Association (IATA), 2009

5.2 Selected Publications

1. Seevers, Robert H. et al. The Use of Mean Kinetic Temperature (MKT) in the Handling, Storage, and Distribution of Temperature Sensitive Pharmaceuticals. *Pharmaceutical Outsourcing.* **2009,** *May/June,* 30-35.
2. *Technical Report No 39 (Revised 2007): Guidance for Temperature Controlled Medicinal Products: Maintaining the Quality of Temperature Sensitive Medicinal Products through the Transportation Environment;* Parenteral Drug Association: Bethesda, MD, 2007.
3. *Technical Report No. 46: Last Mile: Guidance for Good Distribution Practices for Pharmaceutical Products to the End User;* Parenteral Drug Association: Bethesda, MD: 2009.
4. *Technical Report No. 53: Guidance for Industry: Stability Testing to Support Distribution of New Drug Products;* Parenteral Drug Association: Bethesda, MD, 2011.

5.3 Selected Regulatory Authority Websites

1. The International Conference on Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH) www.who.int/en
2. The World Health Organization (WHO) www.who.int/en
3. The European Medicines Agency (EMA) www.ema.europa.eu
4. U.S. Food and Drug Administration (FDA) www.fda.gov
5. Health Canada www.hc-sc.gc.ca
6. U.S. Drug Enforcement Agency www.justice.gov/dea

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